



Robust optical-frequency-comb based on the hybrid mode-locked Er: fiber femtosecond laser

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Abstract: We demonstrate an optical frequency comb in which an Er: fiber-based femtosecond laser employs nonlinear amplifier loop mirror (NALM) and nonlinear polarization evolution (NPE) mode-locking mechanisms. The laser combines advantages of good robustness of NALM and low noise feature of NPE. Our experimental results show that the hybrid mode-locked laser has high power, low relative intensity noise and self-started property, enabling the construction of a robust optical frequency comb system. In-loop relative instabilities of both stabilized repetition rate and carrier-envelope-offset frequency are well below 1×10^{-17} at 1 second integration time.

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1. Introduction

The invention of optical frequency combs has revolutionized optical frequency metrology by phase coherently linking the radio frequency (RF) and optical frequencies [1,2]. These instruments have become irreplaceable in many science and technology applications, such as ultra-stable microwave generation, precision spectroscopy, astronomical spectrograph, high-accuracy time and frequency transfer and optical clocks [3-7]. The femtosecond mode-locked laser [8], which is the core component of an optical frequency comb, determines the performance of the comb. The original work has been focused on the Ti:Sapphire laser because of its low relative intensity noise (RIN) and high repetition rate [9]. However, it is not robust enough for some critical applications, such as space projects, because of its high environmental sensitivity. Consequently, femtosecond fiber lasers have attracted widespread attention because they are compact and widely compatible for long-term operation; they have gradually become a compelling alternative to laser frequency combs [10-14]. The erbium fiber-based laser has especially drawn attention for its combination of low costs and compatibility with components from existing communication networks [15].

To obtain robust erbium fiber lasers that produce high power, high repetition, low RIN, and short pulses with very low free-running timing jitter, various lasers have been designed. All-polarization-maintaining lasers based on a semiconductor saturable absorber mirror (SESAM) and nonlinear amplifier loop mirrors (NALM) are most preferred due to their environmentally stable feature [16]. SESAM mode-locked lasers are the simplest one, which exhibits self-started property with a high repetition rates (above 1 GHz). But SESAM is a slow saturable absorber, leading to a broad pulse accompanied by high frequency noise [17-19]. Alternatively, the designs of the figure-eight (or figure-nine) mode-locked lasers based on the NALM, usually have

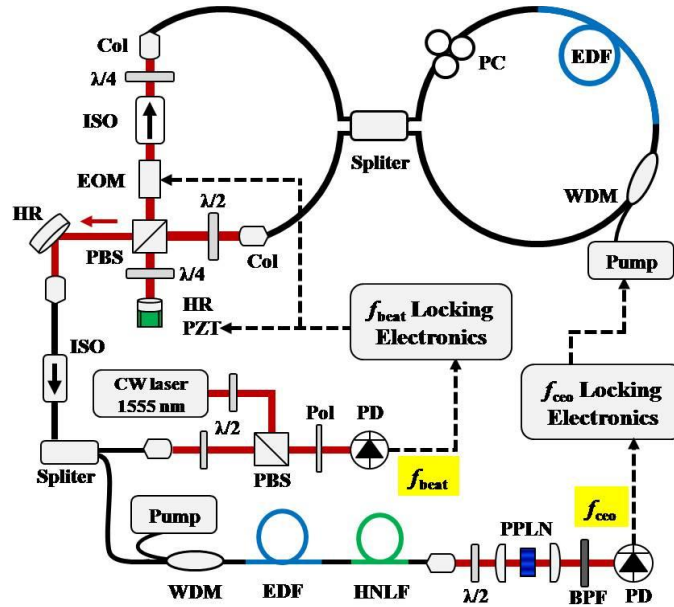


Fig. 1. Experimental setup of the Er-fiber frequency comb based on the hybrid laser. Thick solid lines and curves represent optical fibers; red solid lines represent free-space paths; and dashed lines represent electric signals. Col, collimator; $\lambda/2$, half wave plate; $\lambda/4$, quarter wave plate; ISO, isolator; EOM, electro-optic modulator; PBS, polarization beam splitter; HR, high reflective mirror; PZT, piezo-electric transducer; Splitter, 50 : 50 fiber splitter; WDM, wavelength division multiplexing; EDF, erbium-doped-fiber; PC, polarization controller; Pol, polarizer; PD, photo detector; HNLF, high nonlinear fiber; and BPF, optical band pass filter.

a low fundamental repetition rate (e.g., 50 MHz) and low output power (a few mill watts) [20-24]. Recently, hybrid lasers that insert a SESAM into a nonlinear polarization evolution (NPE) based laser have been reported [25-27]. Hybrid lasers are more robust than the pure NPE mode-locked lasers due to relatively slow saturated absorption effect of the SESAM, and have lower frequency noise level than pure SESAM mode-locked lasers because of the fast and strong saturated absorption effect of the NPE.

In this paper, we demonstrate a comb based on a hybrid scheme for an erbium fiber laser. The erbium fiber laser source incorporates two mode-locking mechanisms: NPE and NALM. NALM is adopted here to initiate mode-locking and shape wide pulses. The ultrashort pulses are obtained in the fast NPE progress by nonlinear filtering of the pulse amplitude. The hybrid mode-locked scheme is more robust than lasers that only use NPE, and the RIN is as low as schemes based on pure NPE. The experimental results also show that the mode-locking state has a large dynamic range and high output power. The repetition rate is up to 168 MHz, and it is controlled by an intra-cavity electro-optic modulator (EOM) and a piezo-electric transducer (PZT). The in-loop instability is approximately 9.4×10^{-18} for an average time of 1 s, and the integrated phase noise is 0.86 rad. Furthermore, a high-quality f_{ceo} of the hybrid laser is measured and phase-locked, and the resulting in-loop frequency instability is about 9.1×10^{-18} with an average time of 1 s. The integrated phase noise is approximately 0.53 rad from 1 Hz to 100 kHz.

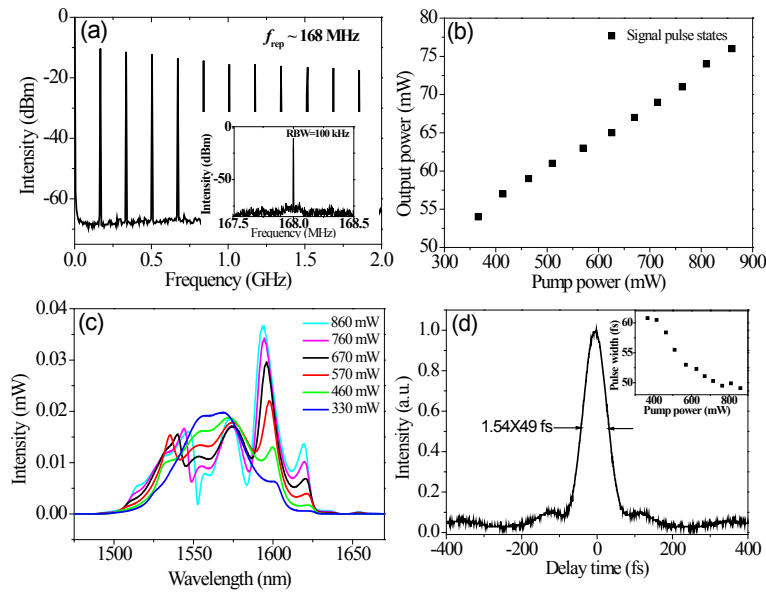


Fig. 2. (a) RF spectrum of the repetition rate of femtosecond pulses, the fine resolution spectrum is shown in the insert. (b) Laser output power as a function of the pump power. (c) Mode-locked laser spectrum. (d) Intensity autocorrelation curve, the pulse width as a function of the pump power is shown in the inset.

2. Experimental setup

The experimental setup of the erbium fiber optical comb is shown in Fig. 1. Two loops are connected by a fiber coupler (50:50): the left one is a primary resonator and the other is a typical NALM. The primary resonator, i.e., the NPE unit, is composed of the SMF-28 fiber and a free-space optical path, including wave plates, a polarization beam splitter (PBS), a polarization independent isolator, a PZT and an EOM. Wave plates are used to optimize the NPE evolution in the cavity, and the isolator makes the laser transmission unidirectional. The EOM is used to control the f_{rep} by changing the optical length. The PZT moving a high reflective mirror is to compensate for long-term drift introduced by temperature changes. The total cavity length is about 122.5 cm, which is composed of 16.5 cm package length of WDM and the splitter, 42 cm gain fiber (Liekki ER110/125) with an absorption of 110 dB/m at 1530 nm, 50 cm SMF-28 fiber pigtail of the WDM, splitter and the collimators, and 14 cm free-space optical path. A polarization controller is employed in the NALM loop to control the phase shift relationship of two counter-propagating incoming light beams by changing the birefringence effect of the fiber in the cavity loop. The hybrid laser is pumped by two 976-nm diode lasers (LC96, Oclaro) with a maximum power of 1 W. The laser cavity dispersion is mainly determined by three components: 50 cm SMF-28 fiber (about -22 fs²/mm), 42 cm Erbium-doped fiber (Liekki 110-4-125, about 12 fs²/mm) and an 8 mm EOM (about 100 fs²/mm) at the central wavelength of -1550 nm; thus, the net dispersion of the cavity is about -5000 fs². After the hybrid laser is mode-locked, the output laser from the PBS is coupled to a collimator for f_{rep} stabilization and f_{ceo} generation.

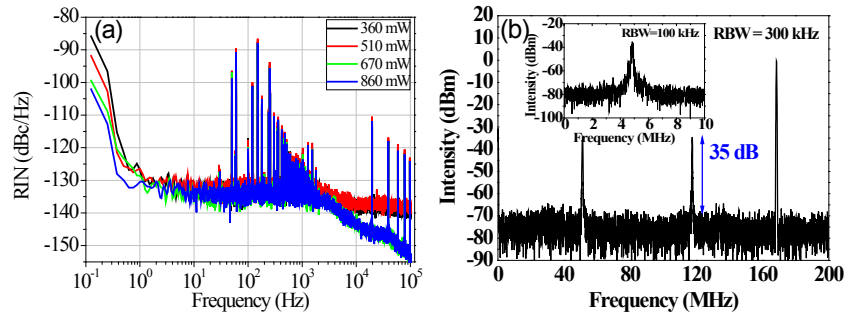


Fig. 3. (a) Residual intensity noise (RIN) of the output pulses. (b) RF spectrum of the free-running f_{ceo} signal.

3. Hybrid mode-locked erbium fiber laser

Mode-locking is easily achieved by adjusting the polarization components, and the laser can self-start mode-locking at pump powers greater than 860 mW. The polarization controller (PC) clamped in NALM is crucial, and mode-locking only occurs when the state of the PC state is proper, which implies that NALM plays an important role. Soliton fission could be observed because of the pulse energy limits in the NALM. Several single pulse states can be found by rotating the half wave plate in the primary resonator or decreasing the pump power, and single pulse operations can be maintained, even when the half wave plate is rotated more than 24° or the pump power is changed from 860 mW to 330 mW. The large mode-locked range guarantees that the laser is insensitive to the polarization drift caused by environmental disturbances. The system can self-start and work normally after being unpowered six months.

Figure 2(a) demonstrates the RF spectrum of the output laser power at a pump power of 860 mW. The fundamental repetition rate is up to 168 MHz, which is much higher than that of a traditional all-fiber figure-eight scheme based on the NALM mechanism. Figure 2(b) demonstrates the laser output as a function of the pump power. The conversion efficiency of the laser varies between 8% and 15% depending on the pump power, and the maximum conversion efficiency is observed at the lowest pump power. Variation of the conversion efficiency may be caused by intensity-dependent nonlinear transmission of the NALM. The maximum pulse energy is 0.75 nJ, and this high pulse energy is beneficial to build an optical comb that requires nonlinear spectral broadening for a self-referenced $f-2f$ interferometer. Figure 2(c) shows the typical mode-locked optical spectra at different pump powers. A series of equivalent width fringes are present in all spectra, which are attributed to the spectral filtering effect of the fiber birefringence of the NALM. When the pump power is gradually weakened, the depth of the fringes diminishes, indicating that the NALM unit of the laser has weak nonlinearity with low pump power. When the pump power is 460 mW, the fringes of the optical spectrum almost disappear, and the corresponding 3-dB spectral width is approximately 75 nm. We also measure the autocorrelation curve shown in Fig. 2(d). The narrowest direct output pulse is approximately 49 fs when a hyperbolic secant pulse is assumed at 860 mW pump power, and the pulse is about 60 fs when the pump power is weakened to 330 mW. These ultrashort pulses are a result of good intra-cavity dispersion compensation.

The RIN of the mode-locked laser plays a key role in amplitude-to-phase conversion of f_{rep} detection and the conversion of intensity noise to f_{ceo} noise in optical combs. We evaluate the RIN of output pulses by coupling about 10 mW into a photodetector (Thorlabs DET10) and then analyzing the output of the photodetector with a fast-Fourier-transformer (SR785). Figure 3(a) demonstrates the double-sideband spectra of the laser with different pump powers. The noise level is approximately -135 dBc/Hz from 10 Hz to 1 kHz, but it rapidly drops above 1 kHz,

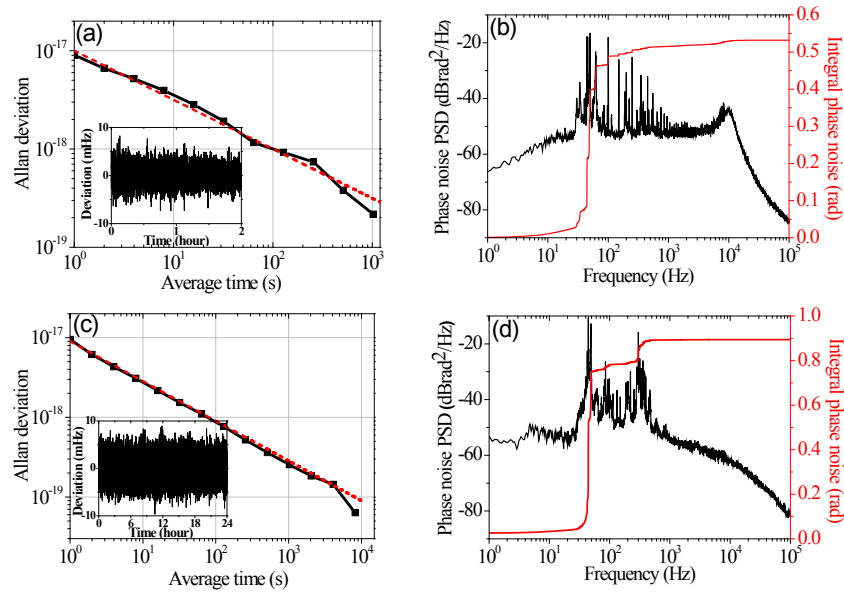


Fig. 4. (a) and (c) are phase noise curves of stabilized f_{ceo} and f_{beat} , black line is phase noise power spectral density (PSD) and red line is integrated phase noise. (b) and (d) are in-loop frequency instabilities of stabilized f_{ceo} and f_{beat} with different average time, the red dash line represents the $\tau^{-1/2}$ dependence due to the dead time of measurement system. Insets are the frequency deviation of the stabilized signal.

especially for the high pump power states, because of the long upper state lifetime of erbium ions. The RIN of our laser is almost the same as that of the NPE mode-locked Yb-fiber laser in [28].

The output of the laser is divided into two parts. One of them is used to detect the f_{ceo} signal with an all-fiber supercontinuum generation and a common-path f - $2f$ configuration [14]; the other is used to stabilize the f_{rep} . The f_{ceo} signal, shown in Fig. 3(b), is produced by using an InGaAs amplified detector. The signal to noise ratio (SNR) is about 35 dB with a resolution bandwidth of 300 kHz, and the estimated linewidth of f_{ceo} is no more than 200 kHz.

4. Phase stabilization of the optical frequency comb

To investigate the long-term stability of the system, f_{ceo} is locked to an external reference by modulating the pump laser current with a homemade loop filter. The coefficient as a function of pump current is about 0.7 MHz/mA. Then, we acquire the locked data with a frequency counter for average time of 1 s. The dead time exists in the measurement system, which degrades the instability for long-terms.

Our results show that f_{ceo} could be stabilized to a setting frequency with a standard deviation of 2 mHz. The in-loop frequency instability, evaluated by the Allan deviation, is approximately 9.0×10^{-18} with 1 s average time, as shown in Fig. 4(a). The slope of $\tau^{-1/2}$ dependence is attributed to the fact that the dead time of frequency counter degrades the coherence of the measurement and converts white phase noise into white frequency noise. Figure 4(b) demonstrates phase noise curves measured with a fast Fourier transform analyzer. The results exhibit integrated phase noise of 0.53 rad from 1 Hz to 100 kHz, and the corresponding time jitter is about 0.44 fs. However, the locking loop still has a relatively low servo bandwidth, which is indicated by the obvious gain bump at about 10 kHz in Fig. 4(b). This limits the vibration-resistance performance, especially the tolerance for high frequency noise. Therefore, the feedback loop

must be improved in subsequent work.

We use both the intra-cavity EOM and PZT to stabilize a comb tooth. The comb tooth at 1555 nm is phase-locked to a 1555-nm narrow linewidth continuous laser. The Allan deviation exhibits in-loop frequency instability of approximately 9.4×10^{-18} with an average time of 1 s, which decreases to 10^{-20} level at 10000 s, as shown in Fig. 4(c). We also measure the phase noise curves, as shown in Fig. 4(d). The corresponding integral phase noise is about 0.86 rad from 1 Hz to 100 kHz, and the time jitter is approximately 0.71 fs.

5. Conclusion

We have demonstrated a phase-stabilized erbium-doped fiber-based frequency comb system based on a hybrid mode-locked femtosecond laser. The laser combines NPE and NALM mechanisms, which enables self-started mode-locking and a low RIN and frequency noise level. The employment of free-space and fiber optics technologies increases the repetition rate to 168 MHz, and the intra-cavity dispersion is well optimized to generate the narrowest pulse of 49 fs. By rotating the wave plates, we obtain a maximum output power of 126 mW. The RIN reach -130 dBc/Hz at 1 Hz and below -145 dBc/Hz for frequencies higher than 10 kHz. This noise level is much lower than that of the reported femtosecond laser based on NALM. We use this laser to construct a frequency comb and investigate the long-term stabilization of the system by stabilizing f_{ceo} and the comb teeth. The in-loop frequency instabilities are below 1×10^{-17} for 1 s average time. The integral phase noises are both less than 1 rad from 1 Hz to 100 kHz, corresponding to residual time jitter of less than 1 fs. The frequency comb based on the hybrid scheme is a good candidate for optical frequency measurements, optical clocks and other applications.

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Disclosures

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